DES-ESOD: DUAL-EVIDENCE SELECTION FOR HIGH-RESOLUTION TINY OBJECT DETECTION

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ABSTRACT

High-resolution UAV imagery makes small-object detection expensive because most spatial regions are background. Selective-computation detectors such as QueryDet, CEASC, and ESOD route computation using a single objectness/ foreground score, but a tiny object's semantic response can collapse toward background level after downsampling, and a region dropped at this stage is unrecoverable by every later network stage. We propose DES-ESOD, an extension of ESOD built around a recall-oriented, dual-evidence region selector that fuses a semantic-objectness branch with a lightweight spectral/local-saliency branch and trains the fused priority with an explicit object-level soft-coverage loss, encouraging at least one selected region to overlap every ground-truth object. Routing reuses ESOD's native fixed-threshold rule unchanged; the realized retained area is reported separately because equal thresholds do not imply equal compute. An ISPP-based decoupled head reduces dense detection-head cost. Evaluation placeholders are retained for UAVDT and TinyPersonV2 (SeaPerson), covering size-bucket recall, AP/AP50, retained area (Occupy), and measured GFLOPs/latency; final cross-dataset results will be inserted after the ongoing experiments finish.

***Index Terms*—** small object detection, selective computation, spectral evidence, coverage-constrained selection, UAV imagery

1. INTRODUCTION

High-resolution aerial and maritime imagery is the regime where small-object detection is hardest: targets occupy a handful of pixels, background dominates the frame, and running a dense detector at full resolution wastes most of its computation on regions with no object of interest [1]. A common remedy is *selective computation*: a lightweight early-stage network scores which spatial regions merit full-resolution processing, and only those regions proceed through the expensive backbone/neck/head [2, 1, 3].

Existing selectors share an implicit assumption: a region containing a real object will produce a sufficiently high foreground/objectness/query response in the lightweight branch. This holds for medium and large objects but becomes unreliable exactly where selective computation matters most – genuinely tiny targets, whose downsampled feature footprint can collapse to one or two activations indistinguishable from background clutter [4]. The two possible selector errors are asymmetric: retaining an extra background region only wastes compute, while dropping a true-object region removes that object from every later stage. An objectness-only selector's failure mode on tiny objects is therefore a silent, unrecoverable false negative, not simply a lower AP.

We ask whether a second, independent evidence signal – local spectral/saliency response, motivated by the observation that tiny targets retain residual high-frequency structure even when their semantic response is weak [4] – can recover objects an objectness-only selector would drop, and whether directly supervising *object-level coverage* (does at least one candidate cell overlap each ground-truth box?) in addition to per-cell classification yields a higher-recall selector under the same routing rule.

We instantiate this idea as DES-ESOD, on ESOD's patch-routing skeleton [1]: a dual-evidence priority head replaces ESOD's single objectness head, trained with an explicit soft object-coverage loss. The results in this paper use ESOD's own native fixed-threshold routing, unchanged – not a fixed top-K cap. This holds the routing rule and threshold fixed, but does not hold realized computation fixed because different score distributions can retain different areas; a fixed-cardinality top-K mode is part of the wider design but is evaluated separately, outside this paper. Because a higher-recall selector naturally retains more image area, we pair it with an efficient decoupled head and report retained area (Occupy) alongside accuracy so this trade-off is never hidden. We define evaluation on two datasets outside ESOD's own reported protocol – UAVDT (vehicles) and TinyPersonV2/SeaPerson (extremely small pedestrians) – to test whether a dual-evidence, coverage-supervised selector transfers across domains it was not tuned on.

Contributions: (1) an object-level soft-coverage loss that directly supervises the event recall actually depends on, instead of a per-cell classification proxy; (2) a lightweight semantic-spectral evidence fusion that repurposes spectral saliency as a routing signal rather than a feature enhancer; (3) an ISPP-based decoupled head that reduces the dense de-

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Table 1. Closest prior work and the gap DES-ESOD targets.

Work	Mechanism	Limitation vs. DES-ESOD
QueryDet [2]	Cascaded sparse query on coarse FPN, ranked by query score	Single objectness-derived score; no evidence for semantically-weak tiny targets; no coverage guarantee
CEASC [3]	Per-layer adaptive activation masking (AMM)	Mask ratio fit per FPN layer as an auxiliary target, not an explicit inference-time budget; single evidence source
ESOD [1]	ObjSeeker heatmap $\rightarrow$ AdaSlicer patch routing $\rightarrow$ Sparse-Head	Pure objectness ObjSeeker; per-cell loss has no guarantee a given GT is covered by any retained patch
SET [4]	Spectral background suppression + saliency enhancement on the full feature map	Spectral evidence enhances representation, not used to estimate per-region miss risk or drive routing
SSABNet [5]	Wasserstein-based Scale-Aware Balancing Loss (SABL) for box regression	Dense detector; improves localization quality, not selective computation
ESOD-YOLO [6]	Shared inverted-residual/partial-conv (ISPP) decoupled head	Despite the name, a dense YOLOv8 detector with no routing or patch-skipping mechanism

tection head's parameter and arithmetic cost while preserving ESOD's prediction contract; and (4) a cross-domain evaluation protocol on UAVDT and TinyPersonV2, with final results retained as placeholders while experiments are in progress.

2. RELATED WORK

Table 1 summarizes the closest prior work and the specific limitation each leaves open relative to DES-ESOD; all six are UAV/aerial or tiny-object detection papers we build directly on or position against.

How DES-ESOD addresses each gap. Against QueryDet/ESOD's single-evidence routing, we add the spectral branch motivated by [4]'s own finding that tiny objects lose semantic salience under downsampling – but where [4] applies spectral processing to enhance the shared feature map, we repurpose the same class of evidence as a second routing signal, fused with the semantic branch before the top-K decision (§3.2). Against CEASC's per-layer mask ratio, our budget is a single explicit top-K applied at inference, not a fitted auxiliary target. Against ESOD's and QueryDet's per-cell objectness supervision, which can be satisfied by many redundant nearby high-score cells without ever guaranteeing that a specific ground-truth object is covered, we replace it with an object-level soft-coverage loss (§3.3) that directly targets the coverage event recall depends on. SSABNet and ESOD-YOLO contribute components, not routing ideas: we adopt SSABNet's SABL as an optional box-regression loss and ESOD-YOLO's ISPP shared stem as our decoupled head (§3.4), both orthogonal to and compatible with the selector contribution above.

3. METHOD

3.1. Overview

DES-ESOD instantiates its recall-oriented, dual-evidence selector on ESOD's patch-routing skeleton (query- and mask-level adapters for other selective-computation backbones are outside this paper's scope). A full-resolution image passes through a shallow stem that produces a spatial priority heatmap. The dual-evidence head (§3.2) scores its cells; ESOD's AdaSlicer thresholds the heatmap and clusters activated cells into routed image patches. The number of resulting patches is input-dependent, not capped. We deliberately keep this rule unchanged for the results in §5: it isolates the dual-evidence priority's score construction from any change to the routing rule itself. DES-ESOD also supports a top-k heatmap-cell mode, which trades this input-dependent activation set for an explicit cell count; that mode is evaluated separately on a development set and is not part of the UAVDT/TinyPersonV2 results reported here. Retained patches alone proceed through the remaining backbone/neck/head; the rest are skipped. Section 3.5 records a planned, not yet executable, within-patch sparse-head adapter.

3.2. Dual-Evidence Priority Head

For candidate cell i and class channel c, with shared shallow feature f_i, a semantic branch predicts the raw logit $z_{i,c}^{\mathrm{sem}} = \mathrm{Conv}_{1\times1}(f_i)$, matching ESOD's ObjSeeker interface. A parallel spectral branch first channel-pools f_i to two maps – per-location max and mean over channels – so the subsequent filters operate on 2 channels regardless of backbone width, then applies a depthwise, per-channel-group 3×3 convolution *initialized* from Laplacian and Sobel-x/Sobel-y kernels (six output channels: one Laplacian and two Sobel responses per pooled input); these kernels are trainable, not fixed, so the network can move away from the classical high-pass initialization where doing so helps routing. A 1×1 stem projects the six filter responses back to f_i's channel width, and a final 1×1 head produces the spectral logit $z_{i,c}^{\mathrm{spec}}$ at the same per-cell resolution. This channel-pooled design (vs. running the same filters on the full, un-pooled channel width) is the low-overhead variant used throughout §5: it is the one structural choice that measurably improved recall on UAVDT (§5) without a second backbone. The two logits are concatenated and passed through a single 1×1 fusion convolution:

$$u_{i,c} = \mathrm{Conv}_{1\times1}\left(\left[z_{i,c}^{\mathrm{sem}} \,\|\, z_{i,c}^{\mathrm{spec}}\right]\right), \qquad s_i = \sigma(u_{i,0}). \quad (1)$$

The released ESOD routing contract consumes channel 0 of the first selector level; the current implementation preserves that contract, so s_i is the class-agnostic routing probability

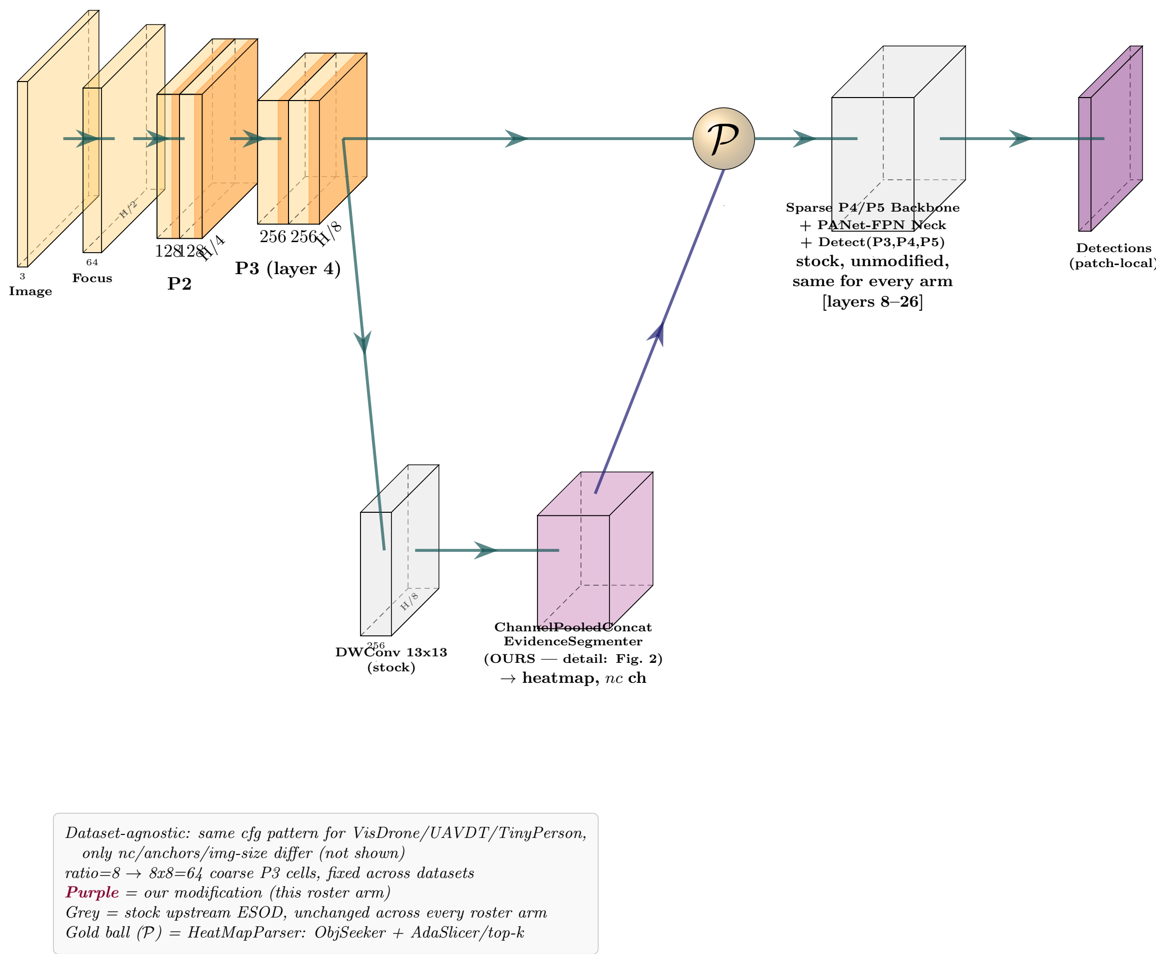

Fig. 1. DES-ESOD system overview. Grey = stock ESOD, unchanged across every selector arm. Purple = the dual-evidence priority head (§3.2) that replaces ESOD's single-evidence ObjSeeker; everything downstream of the HeatMapParser ($\mathcal{P}$) is stock, collapsed into one block since none of it is ours.

trained by the coverage term below, while all fused channels retain pixelwise supervision. For a single-class dataset this distinction disappears. Both branches and the fusion layer add only one extra forward pass on the already-computed shallow feature map; no second backbone is introduced, consistent with the low-overhead constraint that spectral filtering must not itself dominate the compute it is meant to save. Inference routing retains $\mathcal{S} = \{i : s_i > \tau\}$, ESOD's own fixed threshold τ, unchanged (§3.1).

3.3. Object-Level Soft-Coverage Loss

Let $s_i \in [0, 1]$ be candidate cell i's soft routing probability and $\mathcal{N}(j)$ the rectangular set of selector-grid cells whose footprints overlap ground-truth box j. The soft probability that at least one such cell responds to j is

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i), \tag{2}$$

and the coverage loss averages over all ground-truth objects,

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log p_j^{\text{cover}}, \quad w_j = \text{clip}\left(\frac{4}{a_j}, 1, 5\right), \tag{3}$$

where a_j is box area in selector-grid cells. Thus all objects contribute, while objects below four grid cells receive progressively larger weight. As any s_i in $\mathcal{N}(j)$ approaches one, this term approaches zero; it encourages but does not formally guarantee post-threshold coverage. The actual training objective applies both pixelwise BCE and coverage supervision to the fused selector logits:

$$\mathcal{L}_{\text{sel}} = \mathcal{L}_{\text{BCE}}(u, M; \rho{=}2) + 0.5\mathcal{L}_{\text{cover}}, \qquad \mathcal{L} = \mathcal{L}_{\text{det}} + 0.2\mathcal{L}_{\text{sel}}, \tag{4}$$

where M denotes the selector masks and ρ is the positive-class BCE weight. There is no separate semantic-branch auxiliary loss in the current implementation.

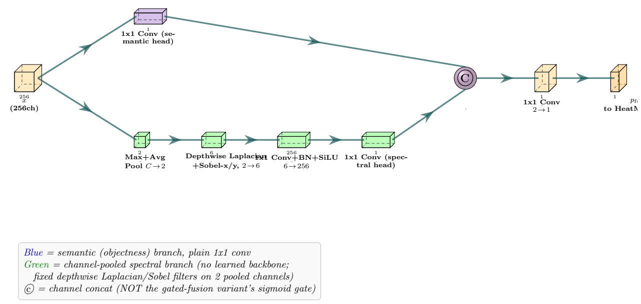

Fig. 2. Dual-evidence priority head. Blue = semantic branch (plain 1×1 conv). Green = channel-pooled spectral branch (no learned backbone). © = channel concatenation.

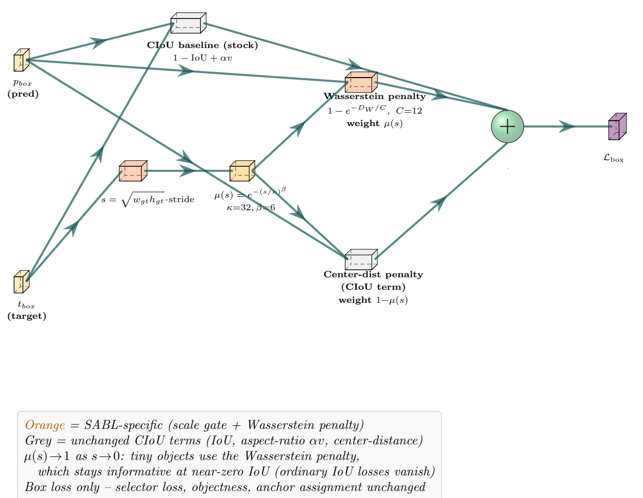

Fig. 3. SABL box loss. Orange = SABL-specific (scale gate $\mu(s)$ + Wasserstein penalty). Grey = unchanged CIoU terms.

3.4. Efficient Decoupled Head

A YOLOv6-style decoupled head is the dominant parameter cost in an ESOD-style detector, large enough to erase a selector's compute savings if left unchanged. We replace it with a shared-stem decoupled head built from ESOD-YOLO's ISPP block [6] (channel-expand 1×1 $\to$ partial 3×3 conv on a quarter of the expanded channels $\to$ project 1×1 with residual), keeping the existing anchor assignment, loss, and decode contract unchanged. Its contribution must still be isolated empirically; the corresponding component-wise ablation is retained as an experiment placeholder.

Box regression optionally uses SSABNet's Scale-Aware Balancing Loss (SABL) [5]. Let $s=\sqrt{w_{gt}h_{gt}}$ be a matched box's geometric-mean scale in input pixels, and D_W the Wasserstein-style distance between the predicted and target Gaussian box representations $(c_x, c_y, w/2, h/2)$, also in input

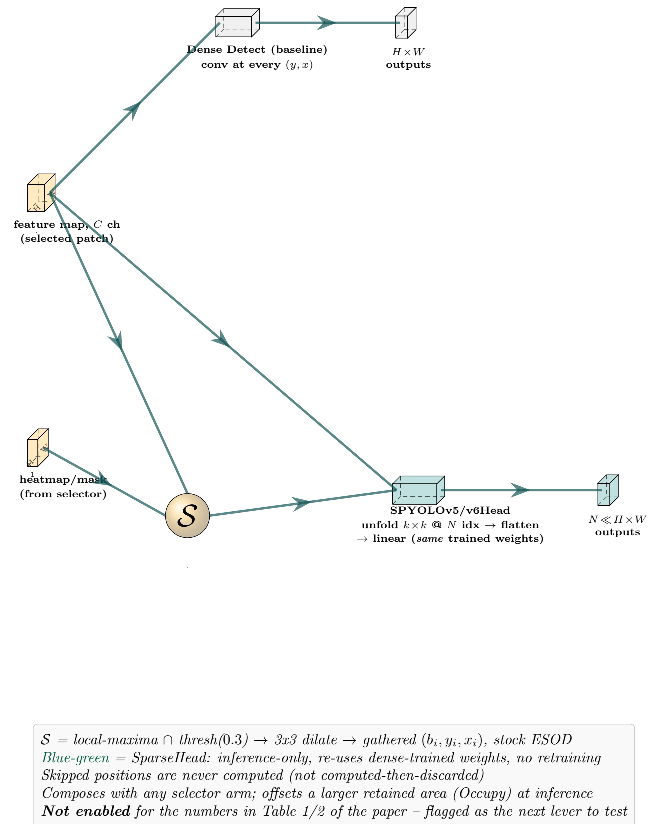

Fig. 4. Planned SparseHead compatibility path. $\mathcal{S}$ = local-maxima ∩ threshold, dilated, as in stock ESOD. The current ISPPHead adapter is not yet implemented.

pixels. A scale-dependent weight

$$\mu(s)=\exp\Big(-(s/\kappa)^\beta\Big),\quad \kappa{=}32,\ \beta{=}6, \tag{5}$$

blends a Wasserstein penalty (dominant for small s) with the normalized squared center-distance penalty ℓ_{ctr} already used by CIoU (dominant for large s):

$$\mathcal{L}_{\text{box}}=1-\text{IoU}+\mu(s)\Big(1-e^{-D_W/C}\Big)+(1-\mu(s))\,\ell_{\text{ctr}}+\alpha v, \tag{6}$$

with C=12 and αv the unchanged CIoU aspect-ratio term. As s shrinks toward zero, $\mu(s)\to1$ and the loss is dominated by a distance metric that stays informative even at near-zero IoU – the regime ordinary IoU-based losses give a vanishing gradient in, which is exactly where tiny objects live. SABL is an orthogonal, opt-in treatment of the regression target, not part of the selector claim.

3.5. SparseHead Compatibility Placeholder

ESOD's SparseHead [1] is a natural candidate for reducing within-patch head computation after a higher-recall selector retains more area. However, the released conversion

Table 2. UAVDT: objectness-only baseline vs. DES-ESOD (dual-evidence selector + SABL + efficient head).

	Baseline	**DES-ESOD**	Δ
mAP@.5	0.344	0.371	+2.7 pp
mAP@.5:.95	0.171	0.195	+2.4 pp
Total recall	84.67%	90.17%	+5.5 pp
Very Tiny recall ($<16\times16$)	83.6%	84.2%	+0.6 pp
Occupy	0.245	0.48	$\sim 2\times$

path currently supports ESOD's convolutional and YOLOv6 heads only; DES-ESOD uses an ISPPHead, for which the corresponding sparse adapter has not yet been implemented. SparseHead is therefore a code and experiment placeholder, not part of the current executable DES-ESOD model or any result reported below. A future adapter must map the ISPP expand, partial-convolution, projection, and prediction layers before the same-weight/no-retraining property can be claimed for DES-ESOD.

4. EXPERIMENTAL SETUP

Datasets. The evaluation protocol covers UAVDT (vehicle detection, three classes: car/truck/bus) and TinyPersonV2/SeaPerson (extremely small pedestrians, single class, whole, un-tiled images only). Neither dataset was used to design or tune DES-ESOD; both are held-out cross-domain checks after the method was developed and ablated on a separate aerial-imagery development set.

Protocol. All arms share a YOLOv5m backbone, matched input resolution, 50 training epochs, SGD with cosine learning-rate scheduling, and identical pretrained initialization. Depending on the arm, the selector head/loss, box loss, and detection head may differ. Component-wise ablations will be inserted here after the ongoing roster completes; until then, the full-model comparison must not be interpreted as isolating one component. Routing uses ESOD's native fixed threshold, unchanged across arms (§3.1); SparseHead (§3.5) is off for every number reported below.

Metrics. We report size-bucket detector recall (Very Tiny $< 16\times16$ px, Tiny 16–32, Small 32–96, Medium/Large $>$ 96 px), COCO-style AP/AP50, measured GFLOPs, batch-1 FPS/latency, and Occupy (fraction of the image area the selector retains) – Occupy is reported specifically because an accuracy gain achieved by retaining more area is a different result from one achieved by selecting the *same* area more accurately.

5. RESULTS

Table 2 shows a consistent win for DES-ESOD over the objectness-only baseline on UAVDT across mAP, total recall, and Very Tiny recall – not a single-axis trade-off. Because both arms use the same unchanged fixed-threshold routing (§3.1), the Occupy row is an honest side effect, not a hidden confound introduced by a favorably-tuned budget: DES-ESOD's score distribution simply clears the threshold at more locations. The open question is how much of the accuracy gain survives once that extra area is paid for. A matched-Occupy or matched-GFLOPs comparison remains an explicit experiment placeholder. SparseHead evaluation additionally depends on the ISPP-compatible adapter described in §3.5.

Table 3. TinyPersonV2 (SeaPerson): roster in progress.

	mAP@.5	**Recall**	**GFLOPs**	**FPS**
Baseline	–	–	–	–
DES-ESOD	–	–	–	–

TinyPersonV2 results (Table 3) are pending at the time of this draft; the data-preparation and six-arm training pipeline are complete and the roster is queued. This table will be populated with audited size-bucket recall and efficiency numbers before submission, following the same protocol as Table 2.

6. CONCLUSION

We presented DES-ESOD, a recall-oriented small-object detector built around a dual-evidence, coverage-supervised region selector that targets the specific failure mode objectness-only selective-computation detectors leave open: silently dropping genuinely tiny, semantically weak objects. On UAVDT, DES-ESOD improves mAP, total recall, and Very Tiny recall over an objectness-only baseline at a larger but not yet budget-matched retained area; a matched-Occupy comparison and the pending TinyPersonV2 cross-domain result are the two remaining pieces needed for a complete, honest claim of generalization.

7. REFERENCES

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